INCH-POUND

A-A-52305
20 August 1992

COMMERCIAL ITEM DESCRIPTION

AUGER, EARTH, BOOM MOUNTED

The General Services Administration has authorized the use of this commercial item description.

1. Abstract. This description covers a commercial model, hydraulically powered, boom mounted earth auger for use on the small emplacement excavator (SEE) or another piece of equipment utilizing the same backhoe configuration as the SEE. The backhoe attachment on the SEE is similar to the commercially available Case 580C backhoe. The hydraulic system on the SEE produces 26 GPM and 2450 psi.

2. Salient characteristics. The earth auger shall be able to bore 12-inch diameter holes, 72 inches deep in soil conditions defined in ASTM D 2487 as: SW, SP, GW, GP, GC, SC, ML, and CL.

2.1 Mounting. The earth auger shall mount in place of SEE backhoe bucket utilizing the mounting bracket as illustrated in figure 1.

2.2 Output shaft. The output shaft which connects the hydraulic motor to the drill bit shall be hexagon shaped and shall utilize a mounting bolt or mounting pin to connect the auger bit to the earth auger motor. The bolt or pin shall be made captive to the earth auger drive unit. The mounting bolt or pin shall be installed and removed utilizing the onboard tools listed in 2.4.

2.3 Drill bit. The auger bit shall be of the double cutting type with replaceable cutting teeth and pilot bits(s). The bit shall be constructed with continuous fluting along the entire length of the drill bit. It shall be able to bore to an effective drilling depth of 72 inches. The bit shall contain a hex shaped connector in order to interface with the motor output shaft.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, US Army Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 3820

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2.4 Cutting teeth. The auger drill bit shall be furnished with replaceable cutting teeth and pilot bit(s). When specified in the contract or purchase order, hard faced tungsten-carbide teeth shall be provided to allow for drilling in rocky or frozen soil. The teeth shall be secured to the auger bit by a positive means which shall prevent the detachment of the teeth from the auger while it is rotating, boring, and retracting. Removal and replacement of teeth shall be accomplished using the following common, onboard hand tools listed below. If additional tools are required, they shall be furnished by the contractor with each auger.

Hammer, 3 pound
Hammer, general purpose, 500 gr
Pliers, combination, general purpose
Screwdriver, combination, general purpose
Wrench, adjustable, 12 inch
Wrench, box type, general purpose, 17x19 mm
Wrench, combination, screwdriver and socket
Wrench, general purpose, 14x15 mm
Wrench, general purpose, 17x19 mm
Wrench, hex key, general purpose
Wrench, open end, general purpose, 8x9 mm
Wrench, open end, general purpose, 10x11 mm
Wrench, open end, general purpose, 12x13 mm
Wrench, open end, general purpose, 19x22 mm
Wrench, open end, general purpose, 24x27 mm
Wrench, open end, general purpose, 30x32 mm

2.5 Hydraulic lines. The pressure and return lines provided with the auger shall be male fittings with port sizes threaded to 3/4-14 NPT. If the hydraulic lines provided are not located on surface A of figure 1, adequate hose shall be provided in order to reach surface A. If whip hoses are provided, they shall be protected against any rubbing that may occur between the hoses and the auger surface metal.

2.6 Operational and storage temperature. The earth auger shall operate satisfactorily when exposed to temperatures ranging from -25 to +120 °F. The earth auger materials of construction and the finishes shall be capable of withstanding prolonged storage in temperatures from -30 to +160 °F. When specified, a quantity of earth augers shall be winterized to permit operation in arctic climates at an ambient temperature of -50 °F and storage at ambient temperatures to -60 °F. The contractor shall furnish one earth auger for examination and demonstration to prove prior to production that his production methods and choice of design detail will produce winterized earth augers that comply with the requirements of this document.

2.7 Noise limits. Noise produced by the auger shall not exceed 90 dB(A) when tested in accordance with SAE J919. Should the noise levels exceed 85 dB(A), then the distance from the earth auger at which a limit of 85 dB(A) is met shall be shown on hazard signs affixed to the earth auger. The hazard signs shall be permanently affixed and shall be clearly visible and legible to all personnel exposed to the hazard. Each hazard sign shall state: "CAUTION, HEARING PROTECTION REQUIRED WITHIN (specify) FEET WHEN EQUIPMENT IS OPERATING".

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2.8 Safety. All rotating or reciprocating, electrically energized, and other parts subject to high operational temperatures, with the exception of the motor output shaft, that are of such a nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. The earth auger shall comply with applicable DOL and OSHA regulations in effect at the time of manufacture. No asbestos or radioactive material shall be used in the auger attachment.

2.9 Treatment and painting. The earth auger shall be treated and painted in accordance with commercial industry's accepted method for retail distribution. The paint color shall conform to color number 34094 of FED-STD-595.

2.10 Identification markings. Identification shall be permanently and legibly marked on a corrosion resisting metal plate securely attached to the earth auger motor in a location inherently protected from damaging conditions. Identification shall include the manufacturer's model and serial number, name, and trademark.

2.11 Materials. The auger shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the attachment may be exposed. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3. Quality assurance provisions.

3.1 Responsibility of inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. A Government representative shall be allowed to witness all contractor testing. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

3.2 Examination. Each earth auger shall be visually examined to determine conformance with all requirements of this description.

3.3 Operational test. When specified, one earth auger shall be subjected to a 100 hour operational test consisting of drilling holes of the required diameter to the required depth in any of the soils mentioned within this description. A test fixture may be used to provide torque input, equal to that found while drilling, in lieu of mounting the auger on a test vehicle and boring holes. During the testing, the earth auger shall experience no failures. A failure is defined as any malfunction which cannot be repaired in 30 minutes by the operator using the tools listed in 2.4 and which causes inability to commence operation, cessation of operation, or performance degradation, or any other unscheduled maintenance action requiring more than one hour to complete. All such malfunctions shall be considered failures except for:

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a. Installation damage.
b. Accident or mishandling.
c. Failure of the test facility or test-peculiar instrumentation.
d. Equipment failures caused by externally applied overstress condition, in excess of the approved test requirement.
e. Normal operating adjustments (nonfailures) specified in the approved equipment operating instructions.
f. Secondary failures within the equipment which are caused directly by relevant or irrelevant primary failures. The secondary failures must be proved to be dependent on the primary failure.
g. Failures caused by human errors.
h. Normal wear out of cutting edges, teeth, or bits. The contractor shall indicate the normal wear out rate prior to testing.

3.6 Contractor certification. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under provisions of the contract.

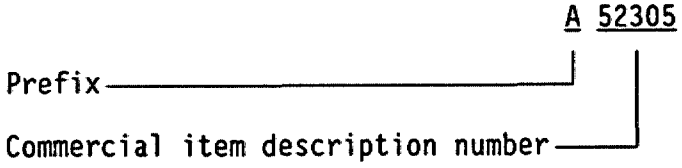
3.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of SAE J916, and all other requirements for this commercial item description are met. The contracting officer has the option of accepting or rejecting the product.

3.8 Regulatory requirements. In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

4. Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or order.

5. CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

5.1 Part or identifying number (PIN). The PIN to be used for the earth auger acquired to this commercial item description is created as follows:



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6. NOTES.

6.1 Referenced documents. Addresses for obtaining copies of referenced documents and color chips are as follows:

ASTM D 2487 should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.

OSHA and DOL documents should be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

SAE J916 and J919 are available by contacting Customer Service, SAE, 400 Commonwealth Drive, Warrendale, PA 15096-0001 (412) 776-4970.

FED-STD-595 is available from the Standardization Documents Ordering Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

Color chip number 34094 of FED-STD-595 may be obtained from the General Services Administration, Room 6662, 7th and D Streets, NW, Washington, DC 20407.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this item description.
b. When hard faced tungsten-carbide cutting teeth shall be provided.
c. When a winterized auger is required.
d. Responsibility for inspection.
e. When an operational test is required.
f. Required preservation, packaging, packing, labeling, and marking.
g. Certification that the earth auger offered meets all requirements of this item description.
h. When commercial manuals with installation and operating instructions and parts list are required.
i. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.

6.3 Sources of supply. The following are suggested sources of supply for this item (competition is not limited to these sources).

a. Eskridge, Inc., 1900 Kansas City Road, P.O. Box 875, Olathe, Kansas 66061.
b. General Equipment Company, P.O. Box 334, 1500 East Main Street, Owatonna, Minnesota 55060.
c. States Engineering Corporation, McMillen Division, 4419 Ardmore Avenue, Fort Wayne, IN 46809-9723
d. Stanley Works, Stanley Hydraulic Tools Division, 3810 S.E. Naef Road, Milwaukee, OR 97267-5698

6.4 Informational drawing. Installation instructions for mounting the earth auger onto the SEE tractor are documented in TA13230E3396.

The preparing activity for this item description is: US Army Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606.

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MILITARY INTERESTS:
COORDINATING ACTIVITY:

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Review activity

Army - AT

CIVIL AGENCY

GSA - FSS

PREPARING ACTIVITY:
Army - ME

Project 3820-0200

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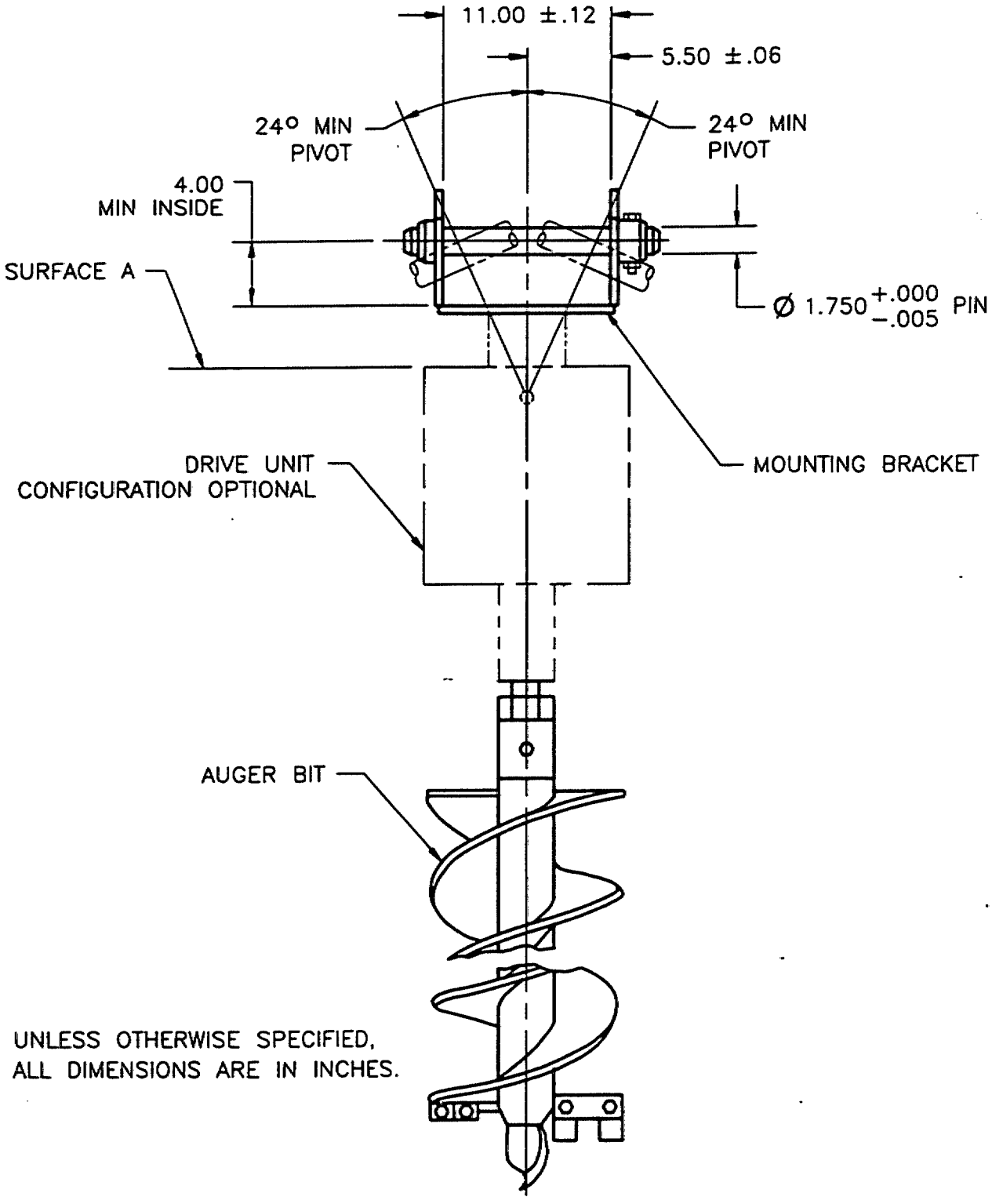


FIGURE 1. Earth auger assembly.

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